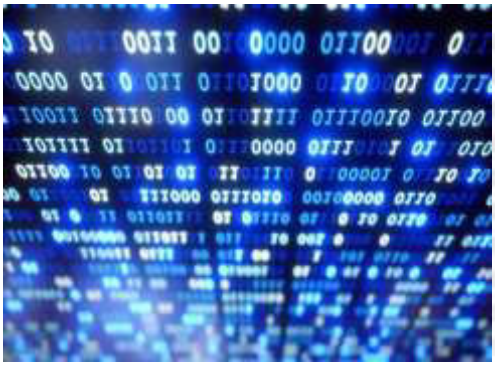
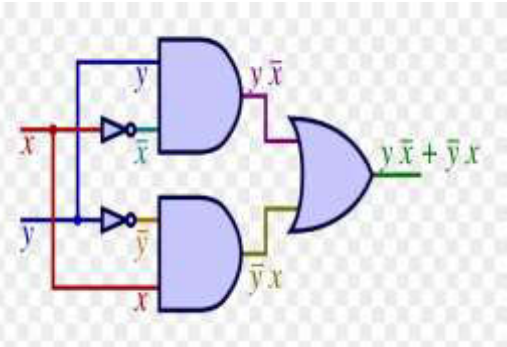
Computer Science

Grade: XI

Number system and conversion Boolean logic





Reference Note

Unit Wise Important Questions for Computer Science XI

Unit 2 Number System and Conversion Boolean Logic

1) What is a number system? What are the types of number systems used in a computer?
2) What is the difference between positional and non-positional number systems?
3) Define radix/base of a number system. What is the radix for binary, octal, decimal and hexadecimal number system?
4) What is a binary number system? Why is it used in computer systems?
5) What is hexadecimal number system?
6) What is Boolean algebra? Give its applications.
7) What is Boolean logic? Explain the three Boolean operators with the help of truth table.
8) What is a truth table? Differentiate between NAND and NOR gate with truth table.
9) What is a logic gate? Construct truth table, logic diagram, Venn diagram of AND, OR and NOT gates.
10) Prove de-Morgan's law with truth table.
11) What is a Boolean law? Explain the different laws of Boolean algebra.
12) State and verify the associative and distributive laws of Boolean algebra.
13) Subtract 10001 from 11011 using 1's and 2's complement method.
14) Convert the following as indicated.
    a) $(AF9)_{16}=(?)_2$ b) $(100101)_2=(?)_{16}$ c) $(129)_{16}=(?)_8$ d) $(2EA)_{16}= (?)_{10}$
15) Write Short notes on: ***a)*** Principle of Duality? b) Use of Boolean algebra

Unit 2 Number System and Conversion Boolean Logic

Computer Number System

The number system plays a vital role in computer calculations. Number system is an organized and systematic way of representing numbers. Number systems are basically of two types: non-positional and positional number systems.

1. Non-Positional Number System

The non-positional number system is a number system in which each symbol represents the same value, regardless of its position in the number. The symbols are simply added to find out the value of a particular number. The most common non-positional number system is the Roman Number System. It is a system of representing numbers devised by the ancient Romans. It is based on certain letters which are given values as numerals.

2. Positional Number System

Positional number system can be represented by a few symbols called digits, which represent different values depending on the position that they occupy. The value of each digit in such a number is determined by the digits itself, the position of the digit in the number, and by the base of the number system. The main positional number systems used in computer are decimal, binary, octal and hexadecimal.

Different types of number system are as follows:

a. **Decimal number system**: A number system having base or radix 10 is called decimal number system.
   Examples: $(789)_{10}$ or $(789)_D$
b. **Binary number system:** A number system having base or radix 2 is called binary number system.
   Examples: $(110)_2$
c. **Octal number system:** A number system having base or radix 6 is called octal number system.
   Examples: $(1234)_8$
d. **Hexadecimal number system**: A number system having base or radix 16 is called hexadecimal number system. Examples: $(C01F)_{16}$ or $(C01F)_H$

1. Decimal Number System

The base or radix of a number system is the number of different symbols available to represent any digit within that system. For example, the decimal system (Base 10) has a radix of 10. Decimal uses different combinations of 10 symbols to represent any value (i.e. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9).

2. Binary Number System

A number system having base or radix 2 is called binary number system. It consists of 2 bits: 0 and 1. It is also known as Binary digits. It is specially used in internal processing of computer system.
An electronic circuit has two states either ON state or OFF state. The bit 1 represents the high voltage i.e. ON state and the bit 0 represents the low voltage i.e. OFF state of an electronic circuit. So it is used in computer system.

3. Octal Number System

A number system having base or radix 8 is called octal number system. It consists of 8 digits: 0, 1, 2, 3, 4, 5, 6, 7). It is also known as octonary number system. The octal system is used in computing as a simple means of expressing binary quantities.

4. **Hexa- Decimal Number System**

A number system having base or radix 16 is called hexadecimal number system. It consists of 16 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F. It is also used in computer basically in memory management.

Number Conversion:

Rules

1. *Decimal Number System to Others (Binary, Octal, Hexa-Decimal) Number System = Divide (2,8,16)*
2. *Others (Binary, Octal, Hexa-Decimal) Number System to Decimal Number System= Multiply (2,8,16)*
3. *Binary Number system to Octal Number system and Vice versa = 3 bit binary Grouping Method (421)*
4. *Binary Number System to Hexa-Decimal Number System and Vice versa= 4 bit binary Grouping Method (8421)*
5. *Octal Number System to Hexa-decimal Number System and Vice versa= 3 bit & 4 bit groping Method (421 & 8421)*

Number Conversion: Rules in short					
1	**Decimal N.S.$_{(10)}$**	**To**	**Others N.S.$_{(2,8,16)}$**	**=**	***Divide (2,8,16)*** ÷
2	**Others N.S. $_{(2,8,16)}$**	**To**	**Decimal N.S.$_{(10)}$**	**=**	***Multiply (2,8,16)*** ***X***
3	**3 bit binary grouping method (421)** → **Octal $_{(8)}$** **Binary $_{(2)}$** ← **4 bit binary grouping method (8421)** → **Hexa- Decimal $_{(16)}$**				

Number Conversion examples

a. Decimal to Binary

Here, $(149)_{10}$

2	**149**	1
2	**74**	0
2	**37**	1
2	**18**	0
2	**9**	1
2	**4**	0
2	**2**	0
2	**1**	1
	0	

Therefore, $(149)_{10} = (10010101)_2$

b. Decimal to Octal

Here, $(804)_{10}$

8	**804**	4
8	**100**	4
8	**12**	4
8	**1**	1
	0	

Therefore, $(804)_{10} = (1444)_8$

c. *Decimal to Hexa- decimal*

Here, $(1600)_{10}$

16	**1600**	0
16	**100**	4
16	**6**	6
	0	

Therefore, $(1600)_{10} = (640)_{16}$

d. *Binary to Decimal*

Here, $(100100)_2 = 2^5 \times 1 + 2^4 \times 0 + 2^3 \times 0 + 2^2 \times 1 + 2^1 \times 0 + 2^0 \times 0$

$= 32 + 0 + 0 + 4 + 0 + 0$

$= (36)_{10}$

e. *Octal to Decimal*

Here, $(2040)_8 = 8^3 \times 2 + 8^2 \times 0 + 8^1 \times 4 + 8^0 \times 0$

$= 1024 + 32$

$= (1056)_{10}$

f. *Hexa-decimal to Decimal*

Here, $(1E0D)_{16} = 16^3 \times 1 + 16^2 \times E + 16^1 \times 0 + 16^0 \times D$

$= 4096 + 256 \times 14 + 0 + 1 \times 13$

$= 4096 + 3584 + 13$

$= (7693)_{10}$

g. *Binary to Octal*

Here, $(110111101)_2$

110	111	101	(: 3 bit binary grouping method
6	7	5	

Therefore, $(110111101)_2 = (675)_8$

h. ***Binary to Hexa-decimal***

Here, $(1001110111)_2$

0010	0111	0111
2	7	7

Therefore, $(1001110111)_2 = (277)_{16}$

i. ***Octal to Binary***

Here, $(375)_8$

3	7	5
011	111	101

Therefore, $(375)_8 = (11111101)_2$

j. ***Hexa-decimal to Binary***

Here, $(ABC)_{16}$

A	B	C
1010	1011	1100

Therefore, $(ABC)_{16} = (101010111100)_2$

k. ***Octal to Hexa-decimal***

Here, $(555)_8$

5	5	5
101	101	101

$= (101101101)_2$

0001	0110	1101
1	6	D

Therefore, $(555)_8 = (16D)_{16}$

l. ***Hexa-decimal to Octal***

Here, $(BCA)_{16}$

B	C	A
1011	1100	1010

$= (101111001010)_{16}$

101	111	001	010
5	7	1	2

Therefore, $(BCA)_{16} = (5712)_8$

Fractional Number System Conversion

a. **Decimal to Binary**

Here, $(0.55)_{10}$

0.55 x 2 = 1.1	1
0.1 x 2 = 0.2	0
0.2 x 2 = 0.4	0
0.4 x 2 = 0.8	0
0.8 x 2 = 1.6	1
0.6 x 2 = 1.2	1

Therefore, $(0.55)_{10} = (0.100011)_2$

b. ***Decimal to Octal***

Here, $(234.997)_{10}$

8	234	2
8	29	5
8	3	3
	0	

Also

0.997 x 8 = 7.976	7
0.976 x 8 = 7.808	7
0.808 x 8 = 6.464	6
0.464 x 8 = 3.712	3
0.712 x 8 = 5.696	5
0.696 x 8 = 5.568	5

Therefore, $(234.997)_{10} = (352.776355)_2$

c. ***Decimal to Hexa-decimal***

Here, $(689.336)_{10}$

16	689	1
16	43	11 = B
16	2	2
	0	

Also

0.336 x 16 = 5.376	5
0.376 x 16 = 6.016	6
0.016 x 16 = 0.256	0
0.256 x 16 = 4.096	4
0.096 x 16 = 1.536	1
0.536 x 16 = 8.576	8

Therefore, $(689.336)_{10} = (2B1.560418)_{16}$

d. ***Binary to Decimal***

Here, $(101.1101)_2$ $= 2^2 \times 1 + 2^1 \times 0 + 2^0 \times 1 + 2^{-1} \times 1 + 2^{-2} \times 1 + 2^{-3} \times 0 + 2^{-4} \times 1$

$= 4 + 0 + 1 + 0.5 + 0.25 + 0 + 0.0625$

$= (5.8125)_{10}$

e. ***Octal to Decimal***

Here, $(0.1042)_8$ $= 8^{-1} \times 1 + 8^{-2} \times 0 + 8^{-3} \times 4 + 8^{-4} \times 2$

$= 0.125 + 0 + 0.0078125 + 0.00048828125$

$= (0.1333)_{10}$

f. ***Hexa-decimal to Decimal***

Here, $(FA.AEF)_{16}$ $= 16^1 \times 15 + 16^0 \times 10 + 16^{-1} \times 10 + 16^{-2} \times 14 + 16^{-3} \times 15$

$= 240 + 10 + 0.625 + 0.0546875 + 0.00366211$

$= 250.68335$

g. ***Binary to Octal***

Here, $(101010.110111)_2$

101	010	110	111
5	2	6	7

Therefore, $(101010.110111)_2 = (52.67)_8$

h. ***Binary to Hexa-decimal***

Here, $(10101.11011)_2$

0001	0101	1101	1000
1	5	D	8

Therefore, $(10101.11011)_2 = (15.D8)_{16}$

i. ***Octal to Binary***

Here, $(77.226)_8$

7	7	2	2	6
111	111	010	010	110

Therefore, $(77.226)_8 = (111111.010010110)_2$

j. ***Octal to Hexa-decimal***

Here, $(0.376)_8$

3	7	6
011	111	110

$= (011111110)_2$

0111	1111
7	F

Therefore, $(0.376)_8 = (0.7F)_{16}$

k. ***Hexa-decimal to Binary***

Here, $(0.5AB)_{16}$

5	A	B
0101	1010	1011

Therefore, $(0.5AB)_{16} = (0.010110101011)_2$

l. ***Hexa-decimal to Octal***

Here, $(0.226)_{16}$

2	2	6
0010	0010	0110

$= (0.001000100110)_2$

001	000	100	110
1	0	4	6

Therefore, $(0.226)_{16} = (0.1046)_8$

Binary Arithmetic

a. ***Binary Addition***

Here, 11111 + 10001

```
    11111
+   10001
   110000
```

Here, 1111 + 1111

```
    1111
+   1111
   11110
```

Rule for addition

0+0=0
0+1=1
1+0=1
1+1=10 (0 with carry over 1)
1+1+1=11 (1 with carry over 1)
1+1+1+1=100 (0 with carry over 10)
1+1+1+1+1=101 (0 with carry over 10)

b. Binary Subtraction

Here, 1100 - 11

```
    1100
-   0011
    1001
```

Here, 101010 - 1001

```
    101010
-   001001
    100001
```

Rule for Subtraction

0-0=0
1-0=1
1-1=0
0-1=1 (with borrowing 1 from left side
Then it becomes 10 i.e. 10-1=1)

c. ***Binary Multiplication***

Here, 1010 x 1010

```
      1010
x     1010
      0000
      1010x
      0000xx
      1010xxx
      1100100
```

Rule for Multiplication

0x0=0
1x0=0
0x1=0
1x1=1

Here, 1111 x 1111

```
        1111
x       1111
        1111
        1111x
        1111xx
        1111xxx
        11100001
```

d. Binary Division

Here, 110100 / 110

```
110) 110100 (1000
     110
     _______
      100
      000
      _______
      100
```

Rule for Division

1÷1=1
0÷1=0
1÷0= not defined
0÷0= not defined

Therefore, Quotient = 1000 and Remainder = 100

Here, 11010001 / 1001

```
1001)    11010001 (10111
         1001
         _________
         1000
         0000
         _________
          10000
          01001
          _________
             1110
             1001
             _____
              1011
              1001
              _____
                10
```

Therefore, Quotient = 10111 and Remainder = 10

1's and 2's Binary Subtraction Method

A compliment is process of representing the negative numbers or bits in digital computer system. Complements are used in digital computers for simplifying the subtraction operation and for logical manipulation. Using complements, all the arithmetic operators can be performed in the form of addition. The process of subtraction using 2's complement is given below:

a. Make the number of digits equal in both minuend and subtrahend.
b. Calculate 2's complement of subtrahend.
c. Calculate sum of minuend and 2's complement of subtrahend.
d. Check the overflow digit (carry).
e. If there is overflow digit, discard it and the remaining bits would be final answer.
f. If there is no overflow bit then the result must be negative. So again, calculate 2's complement of the sum and that would be the final answer.

1'S AND 2'S COMPLEMENT BINARY SUBTRACTION METHOD

1. **Subtract 110 from 1101 using 1's complement binary subtraction method.**

Using 1's complement method:

Here Main value= 1101
Second Value= 110

i. Make the number of bits equal as 1101 and 0110.
ii. 1's complement of second value 0110 is 1001.
iii. Add both the bits:

$$\begin{array}{r} 1101 \\ +\ \underline{1001} \\ \underline{1}\ 0110 \end{array}$$

Here, we got overflow bit so discard it and add the remaining part.
0110 + 1 = 0111
Thus, 1101 - 110 = 111

Using 2's complement method:

Here Main value= 1101
Second Value= 110

i. Make the number of bits equal as 1101 and 0110.
ii. 1's complement of second value 0110 is 1001.
iii. 2's Complement of 1001 + 1 = 1010.
iv. Add both numbers:

$$\begin{array}{r} 1101 \\ +\ \underline{1010} \\ \underline{1}\ 0111 \end{array}$$

Here, we got overflow bit which is discarded and the remaining part is our answer.
Thus, 1101 - 110 = 111

2. **Subtract 1111 from 1100 using 1's and 2's complement binary subtraction method.**

Here, Main value =1100
Second value= 1111
Both are equal digit.

Using 1's complement method:

i. 1's complement of second value 1111 is 0000.
ii. Add both the bits:

$$\begin{array}{r} 1100 \\ +\ \underline{0000} \\ 1100 \end{array}$$

Here, we did not get overflow bit so we again calculate 1's complement of 1100 i.e. 0011 and put minus sign before it.
Thus, 1100 - 1111 = -11

Using 2's complement method:

i. 1's complement of second value 1111 is 0000.
ii. 2's complement of 0000 + 1 = 0001.
iii. Add both numbers:

$$\begin{array}{r} 1100 \\ +\ \underline{0001} \\ 1101 \end{array}$$

Here, we did not get overflow bit so, we again calculate 2's complement of 1101 and put minus sign before it.
i.e. 0010 + 1 =0011
Thus, 1101 - 110 = -11

9'S AND 10'S COMPLEMENT DECIMAL SUBTRACTION METHOD

1. **Subtract 123 from 1234 using 9's and 10's decimal subtraction method.**

Using 9's complement method:

Main value=1234
Second value=123

i. Making the numbers equal in both minuend and subtrahend as 1234 and 0123.
ii. 9's complement of 0123 is (9999-0123) = 9876.
iii. Adding both numbers:

$$\begin{array}{r} 1234 \\ +\ \underline{9876} \\ \underline{1}\ 1110 \end{array}$$

Here, we got overflow digit, so we discard it and add it to the remaining part.
Thus, 1110 + 1 = 1111

Using 10's complement method:

Main value=1234
Second value=123

i. Making the numbers equal in both minuend and subtrahend as 1234 and 0123.
ii. 9's complement of 0123 is (9999-0123) = 9876.
iii. 10's complement of 0123 is (9876+1) = 9877.
iv. Adding both numbers:

$$\begin{array}{r} 1234 \\ +\ \underline{9877} \\ \underline{1}\ 1111 \end{array}$$

Here, we got overflow digit, so we discard it and remaining will the answer.
Thus, 1234 - 123 = 1111

2. **Subtract 4567 from 567 using 9's and 10's decimal subtraction method.**

Using 9's complement method:

Main value= 567
Second value= 4567

Using 9's complement method:

i. Making the numbers equal in both minuend and subtrahend as 0567 and 4567.
ii. 9's complement of 4567 (9999-4567) is 5432.
iii. Adding both numbers:

$$\begin{array}{r} 0567 \\ +\ \underline{5432} \\ 5999 \end{array}$$

Here, we did not get overflow digit, so we again calculate 9's complement of it.
i.e. 9999-5999 becomes 4000
Thus, 567 - 4567 = 4000

Using 10's complement method:

i. Making the numbers equal in both minuend and subtrahend as 0567 and 4567.
ii. 9's complement of 4567 (9999-4567) is 5432.
iii. 10's complement of 4567 (5432+1) is 5433.
iv. Adding both numbers:

$$\begin{array}{r} 0567 \\ +\ \underline{5433} \\ 6000 \end{array}$$

Here, we did not get overflow digit, so we again calculate 10's complement of it.
i.e. (9999-6000)= 3999 + 1 = 4000
Thus, 567 - 4567 = 4000

Boolean algebra

Boolean algebra:
Boolean algebra is a study of mathematical operations performed on certain variables (called binary variables) that can have only two values: true (represented by 1) or false (represented by 0).

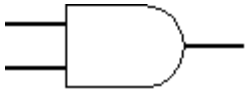
Logic Function (Boolean Function):

A logic function is an expression algebraically with binary variables, logical operation symbols, parenthesis and equal sign, is known as Boolean function. Example, F= A.B.C'+A.B

Truth Table:
Truth table is a table which represents all the possible values of logical variables/statements along with all the possible results of the given combinations of values. For example, following logical statements can have only one of the two values (TRUE (YES) or FALSE (NO)).
Logic gates:
Logic gates perform basic logical functions and are the fundamental building blocks of digital integrated circuits. Most logic gates take an input of two binary values, and output a single value of a 1 or 0.

1. **AND Gate**: AND gate generates true output if all the inputs are true, otherwise it generates false output. It is denoted by (.) operator and graphically represented by:

Logical Symbol

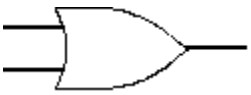


Truth Table

Input		Output
A	**B**	**F=A.B**
0	0	0
0	1	0
1	0	0
1	1	1

2. **OR Gate:** OR gate generates true if at least any one of the input is true, otherwise it generates false Output. It is denoted by (+) operator and graphically represented by:

Logical Symbol

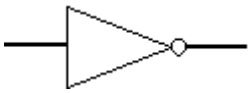


Truth Table

Input		Output
A	**B**	**F=A+B**
0	0	0
0	1	1
1	0	1
1	1	1

3. **NOT Gate:** It is also known as inverter. It inverts the input state from true to false and vice versa. It is denoted by (-) or (') operator and graphically represented by:

Logical Symbol



Truth Table

Input	Output
A	**F=A'**
0	1
1	0

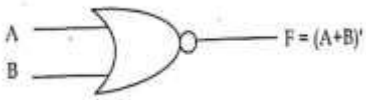
4. **NAND Gate:** NAND gate is the combination of AND and NOT gate. NAND gate generates true (1) output if at least any of the input is false otherwise, it generates false output. Graphically it is represented by:

Logical Symbol

Truth table

Input		Output	
A	**B**	**A.B**	**F=(A.B)'**
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

5. **NOR Gate:** NOR gate is the combination of the OR gate and NOT gate. This electronic gate produces True (1) output when all inputs are False (0) otherwise the output will be False (0). It is the complement of the OR gate. It has two or more inputs and only one output.



Logical Symbol:

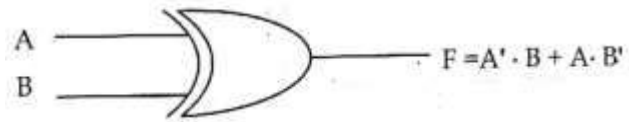
The truth-table of NOR gate is:

Input		Output	
A	**B**	**(A+B)**	**F=(A+B)'**
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

6. **Exclusive (X-OR)Gates:**

The XOR gate produces false output (0) when both the inputs are same otherwise, the output will be true (1). It can also have two or more inputs which produces only one output.

Logical Symbol:



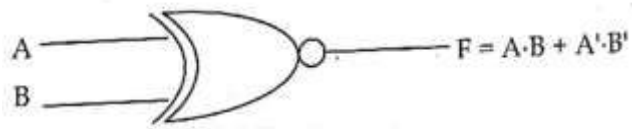
The truth table of X-OR gate is given below:

Inputs						**Output**
A	**B**	**A'**	**B'**	**A'.B**	**A.B'**	**F=A'.B+A.B'**
0	0	1	1	0	0	0
0	1	1	0	1	0	1
1	0	0	1	0	1	1
1	1	0	0	0	0	0

7. **Exclusive-NOR(X-NOR) Gate:**

The XNOR (exclusive-NOR) gate is a combination XOR gate followed by an inverter. Its output is "true" if the inputs are the same and output is "false" if the inputs are different.

The X-NOR gate symbol is given below:



The truth table of X-NOR gate is given below:

Input						**Output**
A	**B**	**A'**	**B'**	**A'.B'**	**A.B**	**F=A.B+A'.B'**
0	0	1	1	1	0	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	0	1	1

There are 2 De Morgan's laws or theorems:

a. Theorem 1: The complement of a sum of variables is equal to the product of the complement of each variables.
   (A+B)' = A'.B'

A	B	A'	B'	A+B	(A+B)'	A'.B'
0	0	1	1	0	1	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	0	0

   Here, (A+B)' = A'.B' thus proved.

b. Theorem 2: The complement of a product of variables is equal to the sum of the complement of each variables.
   (A.B)' = A' + B'

A	B	A'	B'	A.B	(A.B)'	A'+B'
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	0

   Here, (A.B)' = A' + B' thus proved.

Duality principle:

Duality principle state can be obtained by replacing AND (.) with OR (+) and vice versa, 1 with 0 and vice versa keeping variables and complements and variables are unchanged.
For example, duality of the expression A.B' = A+B' and A'.B+C = A'+B.C

Laws of Boolean algebra

The Boolean laws are the rules for manipulating Boolean variables by using Boolean operations. This set of rules states that how variables and values assigned with the Boolean operators reacts. These laws are used to circuit minimization. The process of shorting the length and using minimum components to construct same functioning circuit is called circuit minimization. The Boolean laws are as follows.

7.
a.
Here, A.(B +C) = A+(B.C)

b.
Here, A.B + C.1 + 0.1 = A + B.C + 0.1 + 0

c.
Here, C.D + A.0 + 1 = C + D.A + 1.0

d.
Here, 1.0 + A + C.1 = 0 + 1.A.C + 0

8.
Associative law states that when ORing or ANDing more than two variables, the result is the same regardless of the grouping of the variables.
(a) (A + B) + C = A + (B + C)
(b) (A B) C = A (B C)
Proof:

A	B	C	A+B	B+C	(A+B)+C	A+(B+C)
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	0	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Here, (A + B) + C = A + (B + C) thus proved.

Distributive law states that ORing/ANDing two or more variables and then ANDing/ORing the result with a single variable is equivalent to ANDing/ORing the single variable with each of the two or more variables and then ORing/ANDing the products/sums.
(a) A (B + C) = A.B + A . C
(b) A + (B . C) = (A + B) . (A + C)
Proof:

A	B	C	B+C	A.B	A.C	A.(B+C)	A.B+A.C
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	1	0	1	1	1
1	1	0	1	1	0	1	1
1	1	1	1	1	1	1	1

Here, A.(B+C) = A.B + A.C thus proved.

9.

a.

Here,

AB + A'BC + BC

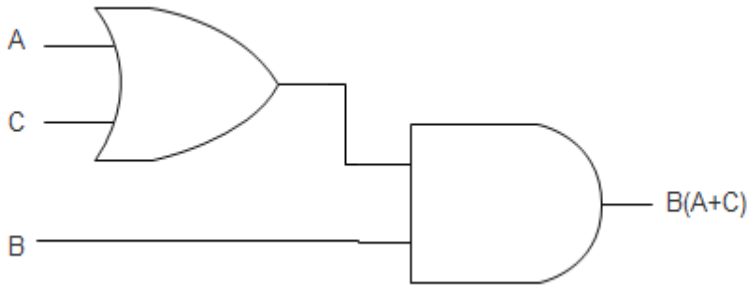
= AB + BC(A'+1)

=AB + BC

=B(A + C)

The truth table is:

A	B	C	A+C	B(A+C)
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	0
1	1	0	1	1
1	1	1	1	1



b.

Here,

PQ' + Q(P + Q) + P(P' + Q)

= PQ' + PQ + QQ + PP' + PQ

= PQ' + PQ + Q + 0

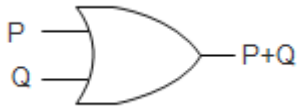
= P(Q'+Q) + Q

= P + Q

The truth table is:

P	Q	P+Q
0	0	0
0	1	1
1	0	1

1	1	1

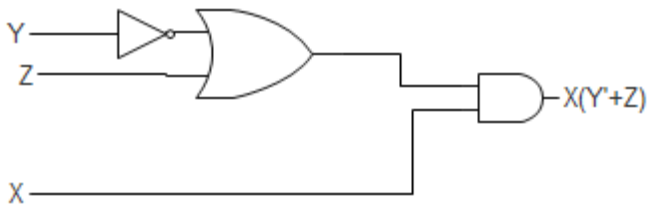


c.
Here,

(X + Y)(XY'Z + XYZ + XY'Z')
= XY'Z + XYZ + XY'Z' + XYY'Z + XYZ + XYY'Z'
= XY'Z + XYZ + XY'Z' + 0 + 0
=XY' (Z + Z') + XYZ
= XY' + XYZ
= X(Y'+YZ)
= X(Y'+ Z)

The truth table is:

X	**Y**	**Z**	**Y'**	**Y'+Z**	**X(Y'+Z)**
0	0	0	1	1	0
0	0	1	1	1	0
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	1	1



10.
Here,
a = A.B.C
b = A' + B + C
c = A'.B'.C

Now, a + b + c = A.B.C + A'+ B + C + A'.B'.C
= B + A' + C + A'B'C
= A' + B' + C

Principle of duality

- Changing to symbol OR (+) operation from AND (.) operation or AND (.) operation from OR (+) operation and digits 0 and 1 that is called principle of duality.
- Example, (x+y).z=(x.y)+z

Uses of gates

- All the gates are used to design digital electronic circuit.
- XOR and XNOR gates are used for circuit minimization.
- The NAND and NOR gates are used for construction of flash memory. Flash memories are used in various devices like PDA, Laptops, mobiles phones etc.
- Flash memory are constructed by using NOR are gates and external memories constructor by using NAND gates. For example, mini SD, Micro SD, Memory Cards.

Uses of Boolean algebra in computer science

- Logical function are useful not only to the hardware designer in implementing circuits but also to software designer in making decision, performing arithmetic, recognizing characters and patterns, checking for errors, formatting output and assembling and disassembling data.